MATH455 - Analysis 4

Abstract Metric, Topological Spaces; Functional Analysis.

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§1 Analysis 4

§1.1 Convolution and Mollifiers

↪**Definition 1.1** (Convolution):

$$(f * g)(x) := \int_{\mathbb{R}^d} f(x-y)g(y)\,\mathrm{d}y = \int_{\mathbb{R}^d} f(y)g(x-y)\,\mathrm{d}y.$$

↪**Proposition 1.1** (Properties of Convolution):

a. $(f * g) * h = f * (g * h)$ (convolution is associative)

b. Let $\tau_z f(x) := f(x-z)$ be the z-translate of x which centers f at z. Then,

$$\tau_z(f * g) = (\tau_z f) * g = f * (\tau_z g).$$

c. $\operatorname{supp}(f * g) \subseteq \overline{\{x + y \mid x \in \operatorname{supp}(f), y \in \operatorname{supp}(g)\}}$.

Proof. (a) Assuming all the necessary integrals are finite, we can change order of integration,

$$\begin{aligned}
((f * g) * h)(x) &= \left(\int f(y)g(x-y)\,\mathrm{d}y\right) * h(x) \\
&= \int \int f(y)g(x-z-y)\,\mathrm{d}y, h(z)\,\mathrm{d}z \\
&= \int \int f(y)g(x-y-z)h(z)\,\mathrm{d}z\,\mathrm{d}y \qquad (y' = x-y) \\
&= \int \int f(x-y')g(y'-z)h(z)\,\mathrm{d}z\,\mathrm{d}y' \\
&= \int f(x-y')(g * h)(y')\,\mathrm{d}y' = (f * (g * h))(x).
\end{aligned}$$

(b) For the first equality,

$$\begin{aligned}
\tau_z(f * g)(x) &= \tau_z \int f(x-y)g(y)\,\mathrm{d}y \\
&= \int f(x-z-y)g(y)\,\mathrm{d}y \\
&= \int (\tau_z f(x-y))g(y)\,\mathrm{d}y = ((\tau_z f) * g)(x).
\end{aligned}$$

The second follows from a change of variables in the second line.

(c) We'll show that $A^c \subseteq (\operatorname{supp}(f * g))^c$ where A the set as defined in the proposition. Let $x \in A^c$, then if $y \in \operatorname{supp}(g)$, $x - y \notin \operatorname{supp}(f)$ so $f(x-y) = 0$; else if $y \notin \operatorname{supp}(g)$ it must be $g(y) = 0$. So, if $x \in A^c$, it must be that

$$\int f(x-y)g(y)\,\mathrm{d}y = \int_{\operatorname{supp}(g)} \underbrace{f(x-y)}_{=0} g(y)\,\mathrm{d}y + \int_{\operatorname{supp}(g)^c} f(x-y)\underbrace{g(y)}_{=0}\,\mathrm{d}y = 0.$$

■

We've been rather loose with finiteness of the convolutions so far. To establish this, we need the following result.

↪**Theorem 1.1** (Young's Inequality): Let $f \in L^1(\mathbb{R}^d), g \in L^p(\mathbb{R}^d)$ for any $p \in [1, \infty]$. Then,
$$\|f * g\|_p \leq \|f\|_1 \|g\|_p,$$
hence $f * g \in L^p(\mathbb{R}^d)$.

Proof. Suppose first $p = \infty$, then
$$(f * g)(x) = \int f(y) g(x-y) \, \mathrm{d}y \leq \|g\|_\infty \int |f(y)| \, \mathrm{d}y = \|g\|_\infty \|f\|_1,$$
for every $x \in \mathbb{R}^d$, so passing to the L^∞-norm,
$$\|f * g\|_\infty \leq \|f\|_1 \|g\|_\infty.$$
Suppose now $p = 1$. Then,
$$\|f * g\|_1 = \int \left| \int f(x-y) g(y) \, \mathrm{d}y \right| \mathrm{d}x.$$
Let $F(x, y) = f(x-y) g(y)$, then for almost every $y \in \mathbb{R}^d$,
$$\begin{aligned} \int |F(x, y)| \, \mathrm{d}x &= \int |g(y)| |f(x-y)| \, \mathrm{d}x \\ &= |g(y)| \int |f(x-y)| \, \mathrm{d}x \\ &= |g(y)| \|f\|_1. \end{aligned}$$
Applying Tonelli's Theorem, we have then
$$\iint |F(x, y)| \, \mathrm{d}y \, \mathrm{d}x = \iint |F(x, y)| \, \mathrm{d}x \, \mathrm{d}y = \int |g(y)| \|f\|_1 \, \mathrm{d}y = \|f\|_1 \|g\|_1,$$
(so really $F \in L^1(\mathbb{R}^d) \times L^1(\mathbb{R}^d)$), hence all together
$$\|f * g\|_1 = \int \left| \int F(x, y) \, \mathrm{d}y \right| \mathrm{d}x \leq \iint |F(x, y)| \, \mathrm{d}y \, \mathrm{d}x = \|f\|_1 \|g\|_1.$$

Remark 1.1: It also follows that for a.e. $x \in \mathbb{R}^d$, $\int |F(x, y)| \, \mathrm{d}y < \infty$, i.e. $\int |f(x - y) g(y)| \, \mathrm{d}y < \infty$. Moreover, since if $g \in L^p(\Omega)$ then $|g|^p \in L^1(\Omega)$, a similar argument gives that for almost every $x \in \mathbb{R}^d$, $\int |f(x-y)| |g(y)|^p \, \mathrm{d}y < \infty$.

Suppose now $1 < p < \infty$. For a.e. $x \in \mathbb{R}^d$, $\int |g(y)|^p |f(x-y)| \, \mathrm{d}y < \infty$, so $g \in L^p(\mathbb{R}^d)$ implies for a.e. $x \in \mathbb{R}^d$, $|g(\cdot)|^p \, |f(x - \cdot)| \in L^1(\mathbb{R}^d)$ as a function of $\cdot$. This further implies $g(y) f^{\frac{1}{p}}(x-y) \in L^p(\mathbb{R}^d, \mathrm{d}y)$. Also, if $f \in L^1(\mathbb{R}^d)$, then $f^{\frac{1}{q}} \in L^q(\mathbb{R}^d)$. All together then,
$$\begin{aligned} \int |f(x-y)| |g(y)| \, \mathrm{d}y &= \int \overbrace{\left| f^{\frac{1}{q}}(x-y) \right|}^{q} \underbrace{\left| f^{\frac{1}{p}}(x-y) \right| |g(y)| \, \mathrm{d}y}_{p} \\ \text{Holder's} \qquad &\leq \left(\int |f(x-y)| \, \mathrm{d}y \right)^{\frac{1}{q}} \left(\int |f(x-y)| |g(y)|^p \, \mathrm{d}y \right)^{\frac{1}{p}}, \end{aligned}$$
hence, raising both sides to the p,

$$|(f*g)(x)|^p \leq \|f\|_1^{\frac{p}{q}} \cdot (|f| * |g|^p)(x)$$

and integrating both sides

$$\int |(f*g)(x)|^p \,\mathrm{d}x \leq \|f\|_1^{\frac{p}{q}} \int \left(\underbrace{|f|}_{\in L^1(\mathbb{R}^d)} * \underbrace{|g|^p}_{\in L^1(\mathbb{R}^d)} \right)(x) \,\mathrm{d}x.$$

Hence, we can bound the right-hand term using the previous case for $p = 1$, and find

$$\begin{aligned} \int |(f*g)(x)|^p \,\mathrm{d}x &\leq \|f\|_1^{\frac{p}{q}} \|f\|_1 \|g^p\|_1 \\ &= \|f\|_1^{\frac{p}{q}+1} \|g\|_p^p \\ &= \|f\|_1^{\frac{p+q}{q}} \|g\|_p^p \\ \left(\frac{p+q}{q} = p\right) \qquad &= \|f\|_1^p \|g\|_p^p, \end{aligned}$$

so raising both sides to $\frac{1}{p}$, we conclude

$$\|f*g\|_p \leq \|f\|_1 \|g\|_p.$$

■

↪**Proposition 1.2**: If $f \in L^1(\mathbb{R}^d)$ and $g \in C^1(\mathbb{R}^d)$ with $\left|\partial_{x_i} g\right| \in L^\infty(\mathbb{R}^d)$ for $i = 1, ..., d$, then $(f*g) \in C^1(\mathbb{R}^d)$ and moreover

$$\partial_{x_i}(f*g) = f * \left(\partial_{x_i} g\right).$$

Remark 1.2: There are many different conditions we can place on f, g to make this true; most basically, we need $|(\partial_i g) * f| < \infty$.

Proof.

$$\frac{\partial}{\partial x_i}\left(\int f(y) g(x-y) \,\mathrm{d}y\right) = \int \underbrace{f(y)}_{\in L^1(\mathbb{R}^d)} \underbrace{\partial_i g(x-y)}_{\in L^\infty(\mathbb{R}^d)} \,\mathrm{d}y < \infty,$$

citing the previous theorem for the finiteness; the dominated convergence theorem allows us to pass the derivative inside. ■

Remark 1.3: This also follows for the gradient; namely $\nabla(f*g) = f*(\nabla g)$ with a component-wise convolution.

Consider the function

$$\rho(x) = \begin{cases} C \exp\left(-\frac{1}{1-|x|^2}\right) & \text{if } |x| \leq 1 \\ 0 & \text{o.w.} \end{cases},$$

where $C = C(d)$ a constant such that $\int_{\mathbb{R}^d} \rho(x)\,\mathrm{d}x = 1$. Then, note that $\rho \in C_c^\infty(\mathbb{R}^d)$ (infinitely differentiable with compact support). Let now

$$\rho_\varepsilon(x) := \frac{1}{\varepsilon^d}\rho\left(\frac{x}{\varepsilon}\right).$$

Notice that $\rho_\varepsilon(x)$ is supported on $B(0, \varepsilon)$, but

$$\int_{\mathbb{R}^d} \rho_\varepsilon(x)\,\mathrm{d}x = \frac{1}{\varepsilon^d}\int_{\mathbb{R}^d} \rho\left(\frac{x}{\varepsilon}\right)\mathrm{d}x = \frac{1}{\varepsilon^d}\cdot\varepsilon^d\cdot\int_{\mathbb{R}^d}\rho(y)\,\mathrm{d}y = 1,$$

for every ε, by making a change of variables $y = \frac{x}{\varepsilon}$. We'll be interested in the convolution

$$f_\varepsilon(x) := (\rho_\varepsilon * f)(x)$$

for some function f. ρ_ε is often called a "convolution kernel". In particular, it is a "good kernel", namely has the properties:

- $\int_{\mathbb{R}^d} \rho_\varepsilon(y)\,\mathrm{d}y = 1$;
- $\int_{\mathbb{R}^d} |\rho_\varepsilon(y)|\,\mathrm{d}y \leq M$ for some finite M;
- $\forall\, \delta > 0$, $\int_{\{|y|>\delta\}} |\rho_\varepsilon(y)|\,\mathrm{d}y \underset{\varepsilon\to 0}{\to} 0$.

The second condition is trivially satisfied in this case since our kernel is nonnegative. The last also follows easily since ρ_ε has compact support; more generally, this imposes rapid decay conditions on the tails of good kernels.

Since $\rho_\varepsilon \in C_c^\infty(\mathbb{R}^d)$, for "reasonable" f, $f_\varepsilon = \rho_\varepsilon * f \in C^\infty(\mathbb{R}^d)$ by the previous proposition. In fact, we'll see that in many contexts $f_\varepsilon \to f$ as $\varepsilon \to 0$ in some notion of convergence. So, f_ε provides a good, now smooth, approximation to f.

↪**Proposition 1.3**: Suppose $f \in L^\infty(\mathbb{R}^d)$ and f_ε is well-defined. Then, if f is continuous at x, then $f_\varepsilon(x) \to f(x)$ as $\varepsilon \to 0$.

If $f \in C(\mathbb{R}^d)$, then $f_\varepsilon \to f$ uniformly on compact sets.

Proof. f continuous at x gives that for every $\eta > 0$ there exists a $\delta > 0$ such that $|f(y) - f(x)| < \eta$ whenver $|x - y| < \delta$. Then

$$\begin{aligned}
|f_\varepsilon(x) - f(x)| &= \left|\int \rho_\varepsilon(y) f(x-y)\,\mathrm{d}y - f(x)\underbrace{\int \rho_\varepsilon(y)\,\mathrm{d}y}_{=1}\right| \\
&= \left|\int \rho_\varepsilon(y)(f(x-y) - f(x))\,\mathrm{d}y\right| \\
&\leq \int_{\{|y|\leq\delta\}} |f(x-y) - f(x)||\rho_\varepsilon(y)|\,\mathrm{d}y + \int_{\{|y|>\delta\}} |f(x-y) - f(x)||\rho_\varepsilon(y)|\,\mathrm{d}y \\
\left(\begin{smallmatrix}\text{cnty in first argument} \\ L^\infty\text{-bound in second}\end{smallmatrix}\right) \quad &\leq \int_{\{|y|\leq\delta\}} \eta|\rho_\varepsilon(y)|\,\mathrm{d}y + 2\|f\|_\infty \int_{|y|>\delta} |\rho_\varepsilon(y)|\,\mathrm{d}y \\
&\leq \eta\cdot M + 2\|f\|_\infty \int_{\{|y|>\delta\}} |\rho_\varepsilon|
\end{aligned}$$

for $\varepsilon \to 0$, by using the second property of good kernels for the first bound. By the last property, the right-most term $\to 0$ as $\varepsilon \to 0$; moreover, then,

$$\lim_{\varepsilon \to 0} |f_\varepsilon(x) - f(x)| \leq C\eta$$

for some C and every $\eta > 0$, and thus $f_\varepsilon(x) \to f(x)$ as $\varepsilon \to 0$.

Now, if $f \in C(\mathbb{R}^d)$ fix a subset $K \subseteq \mathbb{R}^d$ compact. Hence, $\|f\|_{L^\infty(K)} < \infty$ and f uniformly continuous on K since K compact; so the modulus of continuity is uniform for all $x \in K$, so for $\delta > 0$ and for every $x \in K$,

$$\int_{\{|y| \leq \delta\}} |f(x-y) - f(x)||\rho_\varepsilon(y)| \, \mathrm{d}y \leq C\eta.$$

Also, using the bound on f, we may write the second integral in the argument above as

$$\int_{\varepsilon > |y| > \delta} |f(x-y) - f(x)||\rho_\varepsilon(y)| \, \mathrm{d}y \leq \|f\|_{L^\infty(K+B_\varepsilon)} \int_{\{|y| > \delta\}} |\rho_\varepsilon(y)| \, \mathrm{d}y \underset{\varepsilon \to 0}{\to} 0$$

where we take K slighly larger as $K + B_\varepsilon$, which is still compact. So, since this held for all $x \in K$,

$$\max_{x \in K} |f_\varepsilon(x) - f(x)| \underset{\varepsilon \to 0}{\to} 0.$$

Note that we proved the first for general good kernels but the second only in our constructed one. ■

Remark 1.4: This pointwise convergence result is why "good kernels" are called "approximations to the identity".

Remark 1.5: If $f \in C_c(\mathbb{R}^d)$, then $\operatorname{supp}(f_\varepsilon) \subseteq \overline{\operatorname{supp}(f) + B(0, \varepsilon)}$; so, f_ε is compactly supported if f is. Hence in this case $f_\varepsilon \to f$ uniformly on $\mathbb{R}^d$. More generally, there are many different restrictions one can place on the last claim, such as compact support of f, uniform continuity of f, compact support of the kernel, lack of compact support for the kernel but an L^∞ bound on f, etc. In practice, the proofs are all the same, with different bounds; namely one finds something of the form

$$|f_\varepsilon(x) - f(x)| \leq \underbrace{\int_{|y| < \delta} (\ldots)}_{\substack{\text{small by} \\ \text{(uniform) continuity}}} + \underbrace{\int_{|y| \geq \delta} (\ldots)}_{\substack{\text{small by} \\ \text{compact support, etc}}}$$

↪**Theorem 1.2** (Weierstrass Approximation Theorem): Let $[a, b] \subseteq \mathbb{R}$ and let $f \in C([a, b])$. Then for every $\eta > 0$, there exists a polynomial $P_N(x)$ of degree N such that

$$\|P_N - f\|_{L^\infty([a,b])} < \eta.$$

That is, polynomials are dense in $C([a, b])$.

Proof. Extend f to be continuous with compact support on all of $\mathbb{R}$ in whatever convenient way, such that $\operatorname{supp}(f) \subseteq [-M, M]$ for some sufficiently large $M > 0$. Consider now

$$K_\varepsilon(x) := \frac{1}{\sqrt{\varepsilon}} e^{-\frac{\pi x^2}{\varepsilon}},$$

noting that

$$\int_{-\infty}^{\infty} K_\varepsilon(x)\, \mathrm{d}x = \int_{-\infty}^{\infty} \frac{1}{\sqrt{\varepsilon}} e^{-\frac{\pi x^2}{\varepsilon}}\, \mathrm{d}x = 1,$$

which is clear by a change of variables $y = \frac{\sqrt{2\pi}}{\sqrt{\varepsilon}} x$. As a consequence, $\int_{-\infty}^{\infty} |K_\varepsilon(x)|\, \mathrm{d}x = 1 < \infty$, since $K_\varepsilon \geq 0$. Finally,

$$\begin{aligned}
\int_{|x|>\delta} |K_\varepsilon(x)|\, \mathrm{d}x &= \int_{|x|>\delta} \frac{1}{\sqrt{\pi}} e^{-\frac{\pi x^2}{\varepsilon}}\, \mathrm{d}x \\
&= \int_{|y|>\frac{\sqrt{2\pi}}{\sqrt{\varepsilon}}\delta} \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}}\, \mathrm{d}y \\
\text{since } |y| \geq 1 \text{ here for suff. small } \varepsilon \qquad &\leq \int_{|y|>\frac{\sqrt{2\pi}}{\sqrt{\varepsilon}}\delta} \frac{|y|}{\sqrt{2\pi}} \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}}\, \mathrm{d}y \\
&\leq C e^{-\frac{y^2}{2}} \Big|_{\frac{\sqrt{2\pi}}{\sqrt{\varepsilon}}\delta}^{\infty} \underset{\varepsilon \to 0}{\to} 0.
\end{aligned}$$

So, K_ε is a good kernel, and so $(f * K_\varepsilon)(\varepsilon) \underset{\varepsilon \to 0}{\to} f$ uniformly in $[a, b]$ by our last remark. In particular, for $\eta > 0$ there is some $\varepsilon_0 > 0$,

$$\left\| \left(f * K_{\varepsilon_0}\right) - f \right\|_{L^\infty([a,b])} < \frac{\eta}{2}.$$

We claim now that there is a polynomial P_N such that $\left\| P_N - \left(f * K_{\varepsilon_0}\right) \right\|_{L^\infty([a,b])} < \frac{\eta}{2}$. Recall that $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$, which converges uniformly on compact sets. So, there exists a polynomial S_N (from truncating this sum) such that $\left\| K_{\varepsilon_0} - S_N \right\|_{L^\infty([-M,M])} < \frac{\eta}{4\|f\|_\infty M}$. Thus,

$$\begin{aligned}
\left| f * K_{\varepsilon_0}(x) - f * S_n(x) \right| &\leq \left| \int f(x-y)\left(K_{\varepsilon_0}(y) - S_N(y)\right) \mathrm{d}y \right| \\
\operatorname{supp}(f) \subset [-M, M] \qquad &\leq \int_{-M}^{M} |f(x-y)| \left| K_{\varepsilon_0}(y) - S_N(y) \right| \mathrm{d}y \\
&\leq 2M\|f\|_\infty \frac{\eta}{4M\|f\|_\infty} = \frac{\eta}{2},
\end{aligned}$$

for every x. Let $P_N(x) = (f * S_n)(x)$, which we see to be a polynomial. ■

↪**Theorem 1.3**: Let $f \in L^p\left(\mathbb{R}^d\right)$ with $p \in [1, \infty)$. Then $f_\varepsilon \underset{L^p(\mathbb{R}^d)}{\to} f$.

Proof. Since $f \in L^p\left(\mathbb{R}^d\right)$, for every $\eta > 0$ there is a $\tilde{f} \in C_c\left(\mathbb{R}^d\right)$ such that $\left\| f - \tilde{f} \right\|_p < \eta$. Since $\tilde{f} \in C_c\left(\mathbb{R}^d\right)$, by the previous theorem dealing with mollifiers and uniform convergence, $\tilde{f}_\varepsilon \to \tilde{f}$ uniformly. In particular, we have $\left\| \tilde{f}_\varepsilon - \tilde{f} \right\|_p \underset{p}{\to} 0$, hence

$$\|f - f_\varepsilon\|_p \leq \left\| f_\varepsilon - \tilde{f}_\varepsilon \right\|_p + \left\| \tilde{f}_\varepsilon - \tilde{f} \right\|_p + \left\| \tilde{f} - f \right\|_p.$$

We've dealt with the second two bounds. For the first,

$$\begin{aligned} \left\|f_\varepsilon - \tilde{f}_\varepsilon\right\|_p &= \left\|\left(f - \tilde{f}\right) * \rho_\varepsilon\right\|_p \\ \text{(Young's)} \qquad &\leq \|\rho_\varepsilon\|_1 \left\|f - \tilde{f}\right\|_p = \left\|f - \tilde{f}\right\|_p, \end{aligned}$$

so

$$\|f - f_\varepsilon\|_p \leq 2\left\|f - \tilde{f}\right\|_p + \left\|\tilde{f}_\varepsilon - \tilde{f}\right\|_p < 3\eta.$$

■

↪**Corollary 1.1**: $C_c^\infty\left(\mathbb{R}^d\right)$ dense in $L^p\left(\mathbb{R}^d\right)$.

Proof. We showed $\tilde{f}_\varepsilon$ approximates f in $L^p\left(\mathbb{R}^d\right)$, and by construction $\tilde{f}_\varepsilon$ is smooth with compact support. ■